Modular RAG Generation Lifecycle

测试查询：系统如何避免旧数据在重新摄取时提前可见？

The ingestion generation lifecycle uses a generation fence and an active generation pointer.

A worker first claims a lease, writes every chunk into a new staged generation, and publishes that generation only after Chroma, BM25, and image storage all succeed. Queries filter results against the active generation pointer, so stale generations and incomplete writes remain invisible.

A failed worker records the attempt as failed; an expired lease can be reclaimed safely.